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* Matrices. (m×n)

i) 2×2

ii) 3×3

iii) 3×2

iv) 2×3

v) 2×5

vi) 3×1 ← column ~~vector~~ matrice

vii) 1×84 ← ~~row vector~~ row matrice

~~m×n~~ m×n when m = n

n×n ← square matrice

n - square matrice

$$\begin{bmatrix} 1 & 3 & 2 \\ 5 & 4 & 7 \\ 2 & 1 & 0 \end{bmatrix} \leftarrow 3\text{ - square matrice}$$

* Squre matrix

The operations of addition, multiplication, scalar multiplication and transepze can be performed on any n×n matrices would result in an n×n matrix

* Identity matrix

Identity matrix are always occure as square matrix.

For any square matrix;

$$AI = IA = A$$

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

* matrix operations

* matrix addition.

. when we are dealing with the matrix addition ~~row and size~~ row & size need to be equal.

. The resultant matrix also have the same size

(1) $A = \begin{bmatrix} 5 & 2 & 0 \\ 3 & 1 & 3 \end{bmatrix}_{2\times3}$ $B = \begin{bmatrix} 1 & 0 & 5 \\ 6 & -1 & 0 \end{bmatrix}_{2\times3}$ $C = \begin{bmatrix} 1 & 0 \\ 5 & 11 \end{bmatrix}_{2\times2}$

$A+B = \begin{bmatrix} 6 & 2 & 5 \\ 9 & 0 & 3 \end{bmatrix}_{2\times3}$ $A-B = \begin{bmatrix} 4 & 2 & -5 \\ -3 & 2 & 3 \end{bmatrix}_{2\times3}$

$A+C =$ DNE Does not exist

$A-C =$ DNE

* scalar multiplication.

$A = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix}$

$5A = \begin{bmatrix} 5 & 10 \\ 20 & 25 \end{bmatrix}$

. let A be mxn matrix. If we consentrate that a scalar multiplication of matrix aA, the resultant matrix also have the same size of mxn.

Math

* Multiplying two matrices

- matrices are not commutative under multiplication. i.e.

That means

$\boxed{\text{i.e. } A.B \neq B.A}$

- But they are commutative under addition

$\boxed{\text{i.e. } A+B = B+A}$

$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2\times3}$ $B = \begin{bmatrix} 3 & -1 \\ 2 & 0 \\ 1 & 1 \end{bmatrix}_{3\times2}$

$$A \times B = \begin{bmatrix} 3+4+3 & -1+0+3 \\ 12+10+6 & -4+6 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 2 \\ 28 & 2 \end{bmatrix}_{2\times2}$$

(2) $A = \begin{pmatrix} 1, & 2 \end{pmatrix}_{1\times2}$ $B = \begin{bmatrix} 1 & 4 \\ 2 & 1 \end{bmatrix}_{2\times2}$

$$A \times B = \begin{bmatrix} 1+4 & 4+2 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 6 \end{bmatrix}_{1\times2}$$

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 8 & 3 & 4 \end{bmatrix}_{2\times 3} \qquad B = \begin{bmatrix} 1 & -1 \\ -2 & 2 \\ 0 & 1 \end{bmatrix}_{\cancel{2\times}3\times 2}$$

$$A+B = \begin{bmatrix} 3-4 & -3+4+1 \\ 8-6 & -8+6+4 \end{bmatrix}_{2\times 2}$$

$$= \begin{bmatrix} -1 & 2 \\ 2 & 2 \end{bmatrix}_{2\times 2}$$

* Inverse of a 2x2 matrix

$AA^{-1} = I$

$A^{-1} = \frac{1}{|A|}$

~~If A =~~

A square matrix A is said to be invertible or non singular if the following relationship holds.

$$AA^{-1} = A^{-1}A = I$$

$$\boxed{\text{If } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad A^{-1} = \frac{1}{|A|}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}}$$

• If the determinant of that perticular matrix equal to zero, then we can't identify it's inverse

That means the inverse of that perticular matrix is DNE.

determinant

$|A| = ad - cb$

$|A| \neq 0$

① Let $A = \begin{bmatrix} 1 & -1 \\ 2 & -2 \end{bmatrix}_{2\times2}$

identify the inverse of matrix

$D = -2 + 2$

A^{-1} DNE

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(1) let $D = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}$

$D_D = 2 - 2$

$= 0$

D^{-1} DNE

(2) ~~Let D =~~

Let $B = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$

$B^{-1} = \frac{1}{|1+4|} \begin{bmatrix} 1 & -2 \\ 2 & 1 \end{bmatrix}$

$= \begin{bmatrix} \frac{1}{5} & -\frac{2}{5} \\ \frac{2}{5} & \frac{1}{5} \end{bmatrix}$

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Solving simultaneous equations

① $x+y=5$
$x-y=1$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}}_{A} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

with $\begin{bmatrix} x \\ y \end{bmatrix}$ labelled X and $\begin{bmatrix} 5 \\ 1 \end{bmatrix}$ labelled a.

$D_A = (-1-1)$

$= (-2)$

$= 2$

$AX = a$

$A^{-1}AX = A^{-1}a$

$X = A^{-1}a$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{bmatrix}\begin{bmatrix} 5 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{5}{2} + \frac{1}{2} \\ \frac{5}{2} - \frac{1}{2} \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{6}{2} \\ \frac{4}{2} \end{bmatrix}$$

$x = \frac{6}{2}$

$y = \frac{4}{2}$

(3) $-x + 2y = -1$

$x - 2y = 1$

$$\underbrace{\begin{pmatrix} -1 & 2 \\ 1 & -2 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{X} = \underbrace{\begin{pmatrix} -1 \\ 1 \end{pmatrix}}_{a}$$

finding A^{-1}

$D_A = 2 - 2$

$|A| = 0$ DNE

~~Minor~~

* Minor

if $A = \begin{pmatrix} 8 & 3 & 1 \\ 0 & 4 & 5 \\ 2 & 9 & 8 \end{pmatrix}$

$M_{21} = \begin{pmatrix} 8 & 3 & 1 \\ 0 & 4 & 5 \\ 2 & 9 & 8 \end{pmatrix}$

$= \begin{bmatrix} 3 & 1 \\ 9 & 8 \end{bmatrix}$

$M_{23} = \begin{bmatrix} 8 & 3 \\ 2 & 9 \end{bmatrix}$

* Cofactor matrix

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 5 & -1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

$$C_{12} = (-1)^{1+2} M_{12}$$

$$= -1 \begin{bmatrix} 5 & 2 \\ 1 & 3 \end{bmatrix}$$

$$= -1 [15 - 2]$$

$$= -13$$

$$M_{31} = \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$$

$$C_{31} = (-1)^{(3+1)} M_{31}$$

$$= \begin{pmatrix} 1 & 0 \\ -1 & 2 \end{pmatrix}$$

$$C_{31} = 2 + 0$$

$$= 2$$

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(8) find the cofactor matrix of given matrix A

let
$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & -2 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

$$C.M. = \begin{bmatrix} +\begin{vmatrix} -2 & 1 \\ 2 & 2 \end{vmatrix} & -\begin{vmatrix} 0 & 1 \\ 1 & 2 \end{vmatrix} & +\begin{vmatrix} 0 & -2 \\ 1 & 2 \end{vmatrix} \\ -\begin{vmatrix} 0 & 1 \\ 2 & 2 \end{vmatrix} & +\begin{vmatrix} 1 & 1 \\ 1 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} \\ +\begin{vmatrix} 0 & 1 \\ -2 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} & +\begin{vmatrix} 1 & 0 \\ 0 & -2 \end{vmatrix} \end{bmatrix}^T$$

$$= \begin{bmatrix} (-4-2) & -(-1) & (+2) \\ -(-2) & (2-1) & -(2) \\ (+2) & -(1) & (-2) \end{bmatrix}$$

$$= \begin{bmatrix} -6 & +1 & 2 \\ 2 & 1 & -2 \\ 2 & -1 & -2 \end{bmatrix}$$

* Transpose of a matrix

if

$$A = \begin{bmatrix} 1 & 2 & 5 \\ 4 & 3 & 1 \\ 2 & 9 & 7 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 4 & 2 \\ 2 & 3 & 9 \\ 5 & 1 & 7 \end{bmatrix}$$

* Inverse (Adjoint - method)

If A is a square matrix only.

then

$$\cancel{A^{-1} = \frac{A^T}{|A|}}$$

$$\boxed{A^{-1} = \frac{(\text{cofactor matrix})^T}{|A|}}$$

* Determinant of a $n \times n$ matrix

$$\boxed{|A| = \sum_{j=1}^{n} a_{ij} (-1)^{i+j} M_{ij}}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 1 \\ -1 & -2 & 1 \end{bmatrix}$$

$$|A| = \left(-4 \begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix}\right) + \left(0 \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix}\right) + \left(-1 \begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix}\right)$$

$$= -32 + 0 + 0 = -32$$

$$|A| = \left(-2 \begin{vmatrix} 4 & 1 \\ -1 & 1 \end{vmatrix}\right) + 0 + \left(2 \begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix}\right)$$

$$= -10 + 0 + (-22) = -32$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 0 & 1 \\ -1 & -2 & 1 \end{bmatrix}$$

$$C.M. = \begin{bmatrix} +\begin{vmatrix} 0 & 1 \\ -2 & 1 \end{vmatrix} & -\begin{vmatrix} 4 & 1 \\ -1 & 1 \end{vmatrix} & +\begin{vmatrix} 4 & 0 \\ -1 & -2 \end{vmatrix} \\ -\begin{vmatrix} 2 & 3 \\ -2 & 1 \end{vmatrix} & +\begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ -1 & -2 \end{vmatrix} \\ +\begin{vmatrix} 2 & 3 \\ 0 & 1 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 4 & 1 \end{vmatrix} & +\begin{vmatrix} 1 & 2 \\ 4 & 0 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -5 & -8 \\ -8 & 4 & 0 \\ 2 & 11 & -8 \end{bmatrix}$$

$$A^{-1} = \frac{(\text{cofactor matrix})^T}{|A|}$$

$$A^{-1} = \frac{1}{-32} \begin{bmatrix} 2 & -8 & 2 \\ -5 & 4 & 11 \\ -8 & 0 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{16} & \frac{1}{4} & -\frac{1}{16} \\ \frac{5}{32} & -\frac{1}{8} & -\frac{11}{32} \\ \frac{1}{4} & 0 & \frac{1}{4} \end{bmatrix}$$